MODULE 3 BLOOD

Blood.

BLOCK 3 · MODULE 1: COMPOSITION, FORMATION, AND DISORDERS

A reference for the blood video and lab. This chapter covers the composition of blood, plasma and the formed elements, how blood cells are formed, and common blood disorders. The focus is on structure and the named components.

BY THE END

1. Describe the composition of blood: plasma and the formed elements.
2. Identify the structure of erythrocytes, the five leukocytes, and platelets.
3. Explain where and how blood cells are formed.
4. Name common blood disorders and the component each one affects.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Blood

drsrennie-stack.github.io/new-build-bio4-solano/blood-concept-videos.html

Blood, an Overview

Blood is a connective tissue, the body's only liquid tissue. Like every connective tissue it is cells suspended in a matrix; here the matrix, plasma, is a fluid.

COMPOSITION AND PHYSICAL PROPERTIES

Structure	What it is
Blood	A fluid connective tissue that transports materials throughout the body
Plasma	The liquid extracellular matrix of blood, roughly fifty five percent of blood volume
Formed elements	The cells and cell fragments suspended in the plasma, roughly forty five percent of blood volume
Volume	Seven to eight percent of body weight, about five liters in an average adult
Temperature	About thirty eight degrees Celsius, slightly warmer than core body temperature
Viscosity	About five times more viscous than water, from the protein content and the formed elements
Hematocrit	The percentage of blood volume made up of erythrocytes
Buffy coat	The thin pale layer of leukocytes and platelets between the plasma and the erythrocytes in a spun sample
Functions of blood	Transport of gases, nutrients, wastes, and hormones; regulation of temperature and pH; protection through clotting and immune defense

Plasma

Plasma is about ninety two percent water. The rest is dissolved solutes, and the plasma proteins are the most abundant of them. All but the gamma globulins are made in the liver.

THE THREE MAIN PLASMA PROTEINS

Plasma protein	Share	Made by	Role
Albumin	About sixty percent of plasma protein	The liver	The most abundant plasma protein. Maintains the osmotic pressure that holds water in the blood, and carries hormones, fatty acids, and drugs
Globulins	About thirty five percent	The liver and plasma cells	Alpha and beta globulins are transport proteins. Gamma globulins are the antibodies of immune defense
Fibrinogen	About four percent	The liver	The largest plasma protein and the clotting protein. It converts to fibrin threads that form the framework of a clot

THE REST OF PLASMA

Structure	What it is
Water	About ninety two percent of plasma, the solvent that carries every other component
Nutrients	Glucose, amino acids, fatty acids, and vitamins
Wastes	Urea, creatinine, bilirubin, and ammonia
Electrolytes	Sodium, potassium, calcium, chloride, and bicarbonate
Gases	Dissolved oxygen, carbon dioxide, and nitrogen
Serum	Plasma with the clotting proteins removed, the fluid left after blood has clotted

The Formed Elements

The formed elements are the cellular part of blood. There are three kinds, and only one of them is a complete cell type with a nucleus.

COMPARE THEM

Formed element	Also called	Description	Relative number
Erythrocytes	Red blood cells	Biconcave discs without a nucleus, packed with hemoglobin	By far the most numerous
Leukocytes	White blood cells	Complete nucleated cells that defend the body	The least numerous
Platelets	Thrombocytes	Small cell fragments, not whole cells, that act in clotting	Intermediate in number

Erythrocytes

Erythrocytes, the red blood cells, carry oxygen. Almost every feature of their structure is a compromise made to fit more hemoglobin into the cell.

STRUCTURE AND ITS CONSEQUENCES

Structure	What it is
Erythrocyte	A red blood cell, the oxygen carrier of the blood
Biconcave disc	The erythrocyte's shape, a disc thinner at its center. About seven and a half micrometres across, giving thirty percent more surface area than a sphere of the same volume
No nucleus	A mature erythrocyte has ejected its nucleus and most organelles, leaving more room for hemoglobin
Flexible plasma membrane	Lets the disc bend and fold to squeeze single file through the narrowest capillaries
Hemoglobin	The red, iron-containing protein that fills the erythrocyte and binds oxygen for transport. Four subunits, each with a heme group holding one iron
Lifespan	About one hundred and twenty days, after which worn cells are broken down by the spleen and liver

KEY POINT

Every structural oddity of the red cell serves one job. No nucleus means more room for hemoglobin. A biconcave shape means more surface for gas exchange. A flexible membrane means it can fold through a capillary narrower than itself. Structure follows function here more plainly than anywhere else in the course.

Leukocytes

Leukocytes, the white blood cells, defend the body. They sort into granulocytes, which have visible cytoplasmic granules, and agranulocytes, which do not.

THE FIVE TYPES

Leukocyte	Group	Appearance	Defense role
Neutrophil	Granulocyte	A multi-lobed nucleus, three to five lobes, and fine pale granules. The most numerous leukocyte	The first responder to infection. Engulfs and digests bacteria
Eosinophil	Granulocyte	A two-lobed nucleus and coarse red-orange granules	Attacks parasites and moderates allergic reactions
Basophil	Granulocyte	A lobed nucleus hidden by large dark granules. The rarest leukocyte	Releases histamine and other mediators in inflammation
Lymphocyte	Agranulocyte	A large round nucleus filling almost the whole cell	The core of specific immunity. Includes the T cells and B cells
Monocyte	Agranulocyte	The largest leukocyte, with a kidney-shaped nucleus	Leaves the blood and matures into a macrophage that engulfs debris and pathogens

KEY POINT

Never Let Monkeys Eat Bananas: Neutrophils, Lymphocytes, Monocytes, Eosinophils, Basophils, in order from most to least numerous. The order is the answer to the differential count question.

Platelets and Blood Formation

Platelets are not whole cells, and all the formed elements trace back to one stem cell in the red bone marrow. The formation of blood cells is called hematopoiesis.

PLATELETS

Structure	What it is
Platelet	Also called a thrombocyte, a small membrane-bound fragment of cytoplasm, not a complete cell
Megakaryocyte	The giant bone marrow cell whose cytoplasm breaks into thousands of platelets
Role in clotting	Platelets gather and stick at a vessel injury, forming a plug and helping the clotting proteins build a clot

HEMATOPOIESIS

Structure	What it is
Hematopoiesis	The formation of the blood's formed elements
Red bone marrow	The site of hematopoiesis in adults, found in the spongy bone of the axial skeleton and the proximal limb bones
Hemocytoblast	The pluripotent hematopoietic stem cell that every formed element arises from
Myeloid line	The line that gives rise to erythrocytes, platelets, and all the leukocytes except lymphocytes
Lymphoid line	The line that gives rise to the lymphocytes
Reticulocyte	An immature erythrocyte with ribosome remnants still in it. Normally one to two percent of circulating red cells
Erythropoietin	The hormone, made chiefly by the kidney, that stimulates erythrocyte production when tissues are short of oxygen

Blood Disorders

Most blood disorders are a problem of one formed element or one plasma protein, too few, too many, or abnormal.

COMPARE THEM BY THE COMPONENT EACH ONE AFFECTS

Disorder	Component affected	What it is
Anemia	Erythrocytes	Too few erythrocytes or too little hemoglobin, so the blood carries less oxygen
Sickle cell disease	Erythrocytes and hemoglobin	An inherited abnormal hemoglobin distorts erythrocytes into a sickle shape that jams capillaries
Polycythemia	Erythrocytes	An excess of erythrocytes that thickens the blood and slows its flow
Leukemia	Leukocytes	A cancer of white blood cell formation that crowds the marrow with abnormal leukocytes
Leukopenia	Leukocytes	An abnormally low white blood cell count, which weakens the body's defense
Thrombocytopenia	Platelets	An abnormally low platelet count, which raises the risk of bleeding
Hemophilia	Clotting proteins	An inherited deficiency of a clotting factor, so blood clots too slowly

ONE TUBE OF BLOOD, A WHOLE STATUS REPORT

Because blood is a tissue that can be sampled with a needle, a single tube reports on the whole body. Spinning it gives the **hematocrit**, the fraction of red cells; a complete blood count tallies each formed element. A low red cell count points to **anemia**, a high white cell count toward infection or **leukemia**.

The structure of the erythrocyte explains its diseases. In **sickle cell disease** a single abnormal hemoglobin makes the cell rigid and curved, so it cannot fold through a capillary and blocks it instead, causing pain and tissue damage downstream.

Because **hematopoiesis** happens in red bone marrow, a marrow sample drawn from the iliac crest or the sternum is how the blood-forming tissue itself is examined when a count is abnormal.